EE144 Lab #4
Localization using the Particle Filter

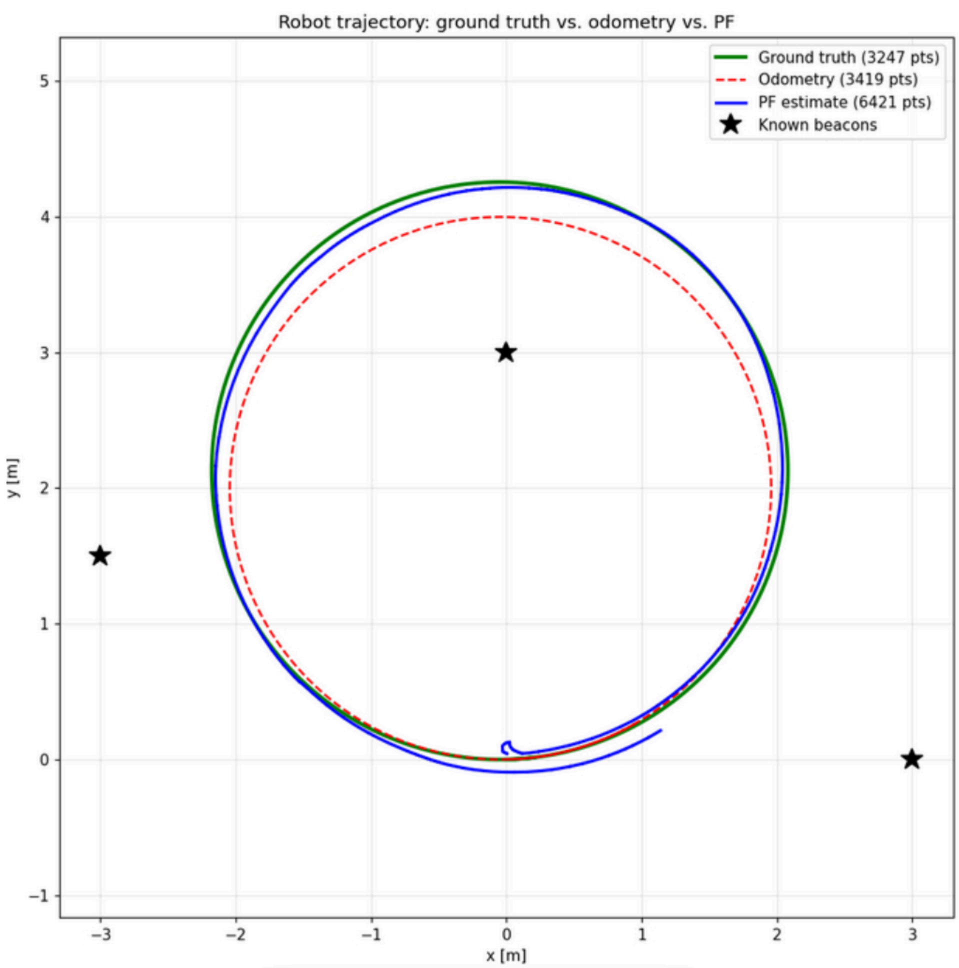
Nathan Taing

Description

In this lab we are working with our third unique filter. It is called a Sampling Importance Resampling Particle Filter. It functions by utilizing a "cloud" of robot states that surround the robot's actual position. It keeps track of these states and associates these possible states with a probability weight based on the information that is received by the sensors. In the update step, the measurements are propagated through the robot's motion model and processed noise is added onto the final result to account for uncertainty. This is through the calculation of measuring the difference between the expected measurement vs the predicted measurements. In order for the weights to remain consistent, they need to be normalized so that they form a probability distribution. We then perform systematic resampling. The bigger weights are more likely to be duplicated and the smaller weights are more likely to be removed. Finally, the estimated position is the weighted mean of all the particles. By repeating this process, the filter continuously refines its estimate and corrects for odometry drift utilizing the measurements. This leads to the most accurate filter we have seen so far.

Results

Green ground Truth, Red Odometry, Blue PF estimate:



Discussion

1) Where does your PF estimate (blue) deviate the most from ground truth (green) on the plot? Speculate on why – is it during the initial convergence, when a beacon is occluded, on a particular part of the circle, or somewhere else?

   The particle filter deviates the most at the initial point as we can see the robot moving towards the center initially, but then self corrects itself to follow ground truth. For most of the run, the particle filter follows the trajectory of ground truth very accurately. The deviation at the beginning can be because of the limited measurement quantity during the early stages of localization. As the filter runs, the cloud of particles converges towards the correct robot state by comparing the expected and actual measurements received. As the filter continues to receive measurements, higher weights are more likely to survive sampling and the particle cloud can converge towards the robot's true position, resulting in a more accurate localization estimate.

2) Look at how the cloud behaves in the first few seconds. Roughly how many seconds did it take for the cloud to collapse from its initial 2 m x 2 m spread down into a tight estimate around the robot? What was the limiting factor – was it the number of beacons visible, the particle count, or something else?

   It took approximately 2 seconds for the cloud to collapse from its initial 2m x 2m spread. The cloud gradually condensed around the robot as the filter received more beacon measurements. The limiting factor was the availability of the beacon measurements. Since the filter requires several observations before it can confidently estimate its own location, it needs more information to correctly weigh the particles to stay in line with our desired path.

3) Compare your PF result to your UKF result from Lab 3 on the same circle. Which one tracks ground truth more tightly? Which one converges faster from a poor initial guess? Briefly explain why the difference makes sense given how each filter represents uncertainty.

   The particle filter tracked the ground truth trajectory more accurately than the UKF. Looking at the graph above, the tracing was near perfect on the green while the UKF's tracing was solid, but not as accurate. The particle filter also recovers from a poor initial guess quicker. This is because the UKF represents uncertainty using only a mean and covariance.The particle filter can represent multiple hypotheses and will gradually eliminate the incorrect ones. This changes as the measurements are received as it has more data to approximate the expected and received data.

4) What happens if your set N_PARTICLES = 50 instead of 500 and re-run? Re-run once with 50 particles and describe the change in plot quality. What does this tell you about the tradeoff between particle count and estimated quality?

   If we lower the amount of particles, the estimate becomes noticeably noisier and deviates from ground truth more often. The fewer the particles, the less accurate the filter becomes and cannot represent the robot's probability distribution as accurately. It is more susceptible to random sampling. This details the tradeoff between particle count and estimated quality. The more particles there are, the more accurately the particle filter can represent the probability distribution, resulting in a higher quality trace of ground truth.